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Aim:

To design JSON representation using JavaScript.

Description:

The webpage is created using various HTML and CSS tags. Below are the tags used along with a short description:

HTML Tags Used:

- `<html>` – The root element of an HTML page.
- `<head>` – Contains metadata, title, styles, and links used in the webpage.
- `<body>` – Contains all the visible content shown on the webpage.
- `<h1>`, `<h2>`, `<h3>`, `<h4>` – Heading tags used to display titles and subheadings.
- `<hr>` – Inserts a horizontal line for separation.
- `<img>` – Used to display images on the webpage.
- `<p>` – Defines a paragraph of text.
- `<a>` – Hyperlink tag used to link to external websites.
- `<label>` – Text associated with form input fields.
- `<input>` – Used to take user input such as text, number, checkbox, radio, and submit.
- `<ul>` – Unordered list to display bullet points.
- `<ol>` – Ordered list to display numbered items.
- `<li>` – List items inside ul or ol.
- `<marquee>` – Used to create scrolling text or images (deprecated tag).
- `<table>` – Used to display data in tabular form.
- `<th>` – Table header cell.
- `<tr>` – Table row.
- `<td>` – Table data cell.
- `<form>` – Used to create a form for user input.
- `<br>` – Line break to shift content to the next line.
- `<style>` – Internal CSS to style HTML elements.

JavaScript Used:

Fetching Data from a JSON File:

- Used the `fetch()` API to load `data.json`.
- Converted the response to a JavaScript object using `.json()`.

DOM Manipulation:

- Used `document.getElementById()` to access HTML elements.
- Created new HTML elements (`<tr>`) dynamically with `document.createElement()`.
- Updated element content using `innerHTML`.
- Appended elements to the DOM with `appendChild()`.

Iteration Over Data:

- Used `forEach()` to loop through arrays (hotels and bookings).
- Event Handling
- Added a submit event listener on the booking form using `addEventListener()`.
- Prevented the default form submission behavior using `event.preventDefault()`.

Form Data Handling:

- Retrieved user input values using `.value` from form input fields.
- Created a JavaScript object to store booking information.

Web Storage (Session Storage):

- Used `sessionStorage.getItem()` to retrieve stored data.
- Used `sessionStorage.setItem()` to save data.
- Parsed JSON strings into objects with `JSON.parse()` and converted objects back to strings with `JSON.stringify()`.

Dynamic Table Rendering:

- Rendered booking data dynamically in a table.
- Re-rendered the table whenever data changed.

Delete Functionality:

- Filtered out items from an array using `Array.filter()`.
- Updated the table and storage after deleting a booking.
- Defined a global function (`window.deleteBooking`) to make the delete button work.

Alert Messages:

- Displayed feedback to the user using `alert()` after a successful booking.

Unique ID Generation:

- Generated unique IDs for bookings using `Date.now()`.

Code:

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hotel Management</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 class="title">Hotel Management System [Ajay Kumar] 2024503701</h1>
  <hr>
```

```

<div class="homeNav">
  <h3><a href="viewHotel.html"> View  Hotels </a></h3>
  <h3><a href="bookRoom.html"> Book  Rooms </a></h3>
  <h3><a href="currentBooking.html"> Current  Booking </a></h3>
</div>
<marquee direction="right" scrollamount="30">
  
  
  
  
</marquee>
<script src="script.js"></script>
</body>
</html>

```

bookRoom.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Hotel Booking</title>
</head>
<body>
  <link rel="stylesheet" href="style.css">
  <h1 class="title">Hotel Room Booking</h1>
  <hr>
  <form id="bookingForm">
    <label>Customer Name</label><br>
    <input type="text" id="name" required><br><br>
    <label>Phone Number</label><br>
    <input type="tel" id="phone" required><br><br>
    <label>Email</label><br>
    <input type="email" id="email" required><br><br>
    <label>Hotel Name</label><br>
    <select id="hotel" required>
      <option value="">Select Hotel</option>
      <option>Sunrise Grand Hotel</option>
      <option>Ocean View Resort</option>
      <option>Mountain Stay Inn</option>
      <option>Royal Palace Hotel</option>
    </select><br><br>
    <label>Check-in Date</label><br>
    <input type="date" id="checkin" required><br><br>
    <label>Check-out Date</label><br>
    <input type="date" id="checkout" required><br><br>
    <label>Number of Guests</label><br>

```

```
<input type="number" id="guests" min="1" max="10" required><br><br>
<button type="submit">Book Now</button>
</form>
<h3><a href="currentBooking.html"> Current Booking </a></h3>
<script src="script.js"></script>
</body>
</html>
```

currentBooking.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Current Booking</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 class="title">Currnet Booking Details</h1>
  <hr>
  <table border="1" class="viewCss">
    <thead>
      <tr>
        <th>Name</th>
        <th>Phone</th>
        <th>Email</th>
        <th>Hotel</th>
        <th>Check-in</th>
        <th>Check-out</th>
        <th>Guests</th>
        <th>Action</th>
      </tr>
    </thead>
    <tbody id="bookingTable">
    </tbody>
  </table>
  <script src="script.js"></script>
</body>
</html>
```

viewHotel.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Hotel Data</title>
  <link rel="stylesheet" href="style.css">
```

```

</head>
<body>
  <h1 class="title">View Hotels Details</h1>
  <hr>
  <table border="1" class="viewCss">
    <thead>
      <tr class="tr">
        <th>HotelId</th>
        <th>Name</th>
        <th>Location</th>
        <th>Contact</th>
        <th>Ratings</th>
      </tr>
    </thead>
    <tbody id="hotelTable">
    </tbody>
  </table>
  <script src="script.js"></script>
</body>
</html>

```

Script.js:

```

fetch('data.json')
  .then(response => response.json())
  .then(data=>{
    const table = document.getElementById('hotelTable');
    data.hotels.forEach(element => {
      const row = document.createElement('tr');
      row.innerHTML = `
        <td>${element.hotelId}</td>
        <td>${element.name}</td>
        <td>${element.location.address} , ${element.location.city},${element.location.country}</td>
        <td>${element.contact.phone}, ${element.contact.email}</td>
        <td>${element.rating}</td>
      `;
      table.appendChild(row);
    });
  })
  .catch(err=>console.log("Error : .err"));
const bookingForm = document.getElementById("bookingForm");
if (bookingForm) {
  bookingForm.addEventListener("submit", function(e) {
    e.preventDefault();
    const bookingData = {
      id: Date.now(),
      name: document.getElementById("name").value,
      phone: document.getElementById("phone").value,

```

```

    email: document.getElementById("email").value,
    hotel: document.getElementById("hotel").value,
    checkIn: document.getElementById("checkin").value,
    checkOut: document.getElementById("checkout").value,
    guests: document.getElementById("guests").value
  };
  let bookings = JSON.parse(sessionStorage.getItem("bookings")) || [];
  bookings.push(bookingData);
  sessionStorage.setItem("bookings", JSON.stringify(bookings));
  alert("Booking added successfully!");
  bookingForm.reset();
});
}
const bookingTable = document.getElementById("bookingTable");
if (bookingTable) {

  let bookings = JSON.parse(sessionStorage.getItem("bookings")) || [];
  function renderTable() {
    bookingTable.innerHTML = "";
    bookings.forEach((booking) => {
      const row = document.createElement("tr");
      row.innerHTML = `
        <td>${booking.name}</td>
        <td>${booking.phone}</td>
        <td>${booking.email}</td>
        <td>${booking.hotel}</td>
        <td>${booking.checkIn}</td>
        <td>${booking.checkOut}</td>
        <td>${booking.guests}</td>
        <td><button onclick="deleteBooking(${booking.id})">Delete</button></td>
      `;
      bookingTable.appendChild(row);
    });
  }
  renderTable();
  // Function to delete booking
  window.deleteBooking = function(id) {
    bookings = bookings.filter(b => b.id !== id);
    sessionStorage.setItem("bookings", JSON.stringify(bookings));
    renderTable();
  };
}
Data.json:
{
  "hotels": [
    {

```

```
"hotelId": "H1001",
"name": "Sunrise Grand Hotel",
"location": {
  "address": "123 Beach Road",
  "city": "Miami",
  "country": "USA"
},
"contact": {
  "phone": "+1-555-123-4567",
  "email": "info@sunrisegrand.com"
},
"rating": 4.5
},
{
  "hotelId": "H1002",
  "name": "Ocean View Resort",
  "location": {
    "address": "45 Ocean Drive",
    "city": "Los Angeles",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-234-5678",
    "email": "contact@oceanview.com"
  },
  "rating": 4.2
},
{
  "hotelId": "H1003",
  "name": "Mountain Stay Inn",
  "location": {
    "address": "78 Hill Road",
    "city": "Denver",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-345-6789",
    "email": "support@mountainstay.com"
  },
  "rating": 4.0
},
{
  "hotelId": "H1004",
  "name": "Royal Palace Hotel",
  "location": {
    "address": "12 King Street",
```

```
"city": "London",
"country": "UK"
},
"contact": {
  "phone": "+44-555-456-7890",
  "email": "info@royalpalace.co.uk"
},
"rating": 4.7
},
{
  "hotelId": "H1005",
  "name": "City Lights Hotel",
  "location": {
    "address": "88 Downtown Ave",
    "city": "New York",
    "country": "USA"
  },
  "contact": {
    "phone": "+1-555-567-8901",
    "email": "hello@citylights.com"
  },
  "rating": 4.3
}
}]}
```

Output:

Hotel Management System [Ajay Kumar] 2024503701

[View Hotels](#)

[Book Rooms](#)

[Current Booking](#)



Figure(a) Dashboard Page

View Hotels Details

HotelId	Name	Location	Contact	Ratings
H1001	Sunrise Grand Hotel	123 Beach Road , Miami,USA	+1-555-123-4567, info@sunrisegrand.com	4.5
H1002	Ocean View Resort	45 Ocean Drive , Los Angeles,USA	+1-555-234-5678, contact@oceanview.com	4.2
H1003	Mountain Stay Inn	78 Hill Road , Denver,USA	+1-555-345-6789, support@mountainstay.com	4
H1004	Royal Palace Hotel	12 King Street , London,UK	+44-555-456-7890, info@royalpalace.co.uk	4.7
H1005	City Lights Hotel	88 Downtown Ave , New York,USA	+1-555-567-8901, hello@citylights.com	4.3
H1006	Desert Oasis Resort	7 Palm Road , Dubai,UAE	+971-555-678-9012, contact@desertoasis.ae	4.8
H1007	Lakeview Comfort Inn	56 Lake Road , Zurich,Switzerland	+41-555-789-0123, info@lakeview.ch	4.4
H1008	Green Valley Lodge	90 Forest Lane , Vancouver,Canada	+1-555-890-1234, support@greenvalley.ca	4.1
H1009	Golden Sands Resort	21 Shoreline Blvd , Goa,India	+91-555-901-2345, info@goldensands.in	4.6
H1010	Skyline Business Hotel	300 Corporate Park , Singapore,Singapore	+65-555-012-3456, business@skyline.sg	4.4
H1011	Heritage Comfort Stay	15 Old Town Road , Jaipur,India	+91-555-123-7890, contact@heritagestay.in	4.2

Figure(b): Data fetched from json file

127.0.0.1:5500 says
Booking added successfully!
OK

Phone Number

Email

Hotel Name

Check-in Date

Check-out Date

Number of Guests

Book Now

Current Booking

Figure(c): Adding the Customer's data dynamically

Currnet Booking Details

Name	Phone	Email	Hotel	Check-in	Check-out	Guests	Action
aji	9638527410	aji@gmail.com	Ocean View Resort	2026-01-06	2026-01-08	5	<button>Delete</button>
sanj	98765643210	san@gmail.com	Sunrise Grand Hotel	2026-01-07	2026-01-30	2	<button>Delete</button>
kumar	8523699741	kuma@gmail.com	Ocean View Resort	2026-01-14	2026-01-30	5	<button>Delete</button>

Figure(d): Viewing and deleting the customer data dynamically

Result:

Thus, the JSON representation was successfully completed using JavaScript, demonstrating basic data handling and manipulation by building Hotel management system.